Weil Conjectures

Part: Appendices

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Appendices

To define $Rf_!$ in Part I we must compactify every separated morphism of finite type, and to prove proper base change we must dominate every proper morphism by a projective one. These are Nagata's theorem and Chow's lemma (Appendix A*). Nagata's theorem, followed by normalization, also gives the smooth compactification of a curve used in Part III (Lemma 14.1*).

Proper base change ends with a change of separably closed ground field. There we need the cohomology of curves for all torsion coefficients, including torsion divisible by the characteristic (Theorem 4.20*). Purity starts from sheaves of rank one. Their geometric monodromy is finite because Frobenius fixes no class in H^1 of a curve (Proposition 14.3*). For coefficients prime to the characteristic, both facts come from the Jacobian. Indeed, $H^1(C, \mu_n)$ is the n-torsion of the Jacobian, and a fixed class would give ℓ-power torsion points of unbounded order among its finitely many rational points. Appendices Appendix B*, Appendix C* and Appendix D* prove both facts.

Rankin's argument in Chapter 14* rests on the determinant weights of tensor and exterior powers (Proposition 14.19*). These come from the structure of the monodromy group (Lemma 14.15*), and Appendix F* proves the facts about algebraic groups over $\overline{\mathbf{Q}}_\ell$ behind it.

Appendix E* is of a different kind. Following Varshavsky [Var05], it computes the local terms of contracting correspondences in a second way; Chapter 12* does not depend on it.

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References marked * are to parts of the blueprint this document leaves out:

Theorem 4.20 Cohomology of curves, in I The six operations (`theorem-cohomology-curves`)
Chapter 12 The trace formula, in II Duality and the trace formula (`chap-The-trace-formula`)
Chapter 14 Purity of real sheaves, in III Weights and the Riemann hypothesis (`chap-Purity-of-real-sheaves`)
Lemma 14.1 A proper smooth model, in III Weights and the Riemann hypothesis (`lemma-proper-smooth-model`)
Proposition 14.3 No fixed class in degree one, in III Weights and the Riemann hypothesis (`proposition-no-fixed-class-degree-one`)
Lemma 14.15 The identity component, in III Weights and the Riemann hypothesis (`lemma-identity-component`)
Proposition 14.19 Determinant weights of tensor operations, in III Weights and the Riemann hypothesis (`proposition-determinant-weights-tensor-operations`)
Appendix A Nagata compactification and Chow's lemma, in Appendices (`chap-Nagata-compactification-and-Chows-lemma`)
Appendix B Abelian varieties, in Appendices (`chap-Abelian-varieties`)
Appendix C The Picard scheme of a curve, in Appendices (`chap-The-Picard-scheme-of-a-curve`)
Appendix D Cohomology of torsion sheaves on curves, in Appendices (`chap-Cohomology-of-torsion-sheaves-on-curves`)
Appendix E Specialization and contracting correspondences, in Appendices (`chap-Specialization-and-contracting-correspondences`)
Appendix F Algebraic groups in characteristic zero, in Appendices (`chap-Algebraic-groups-in-characteristic-zero`)

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